

PRITHVEX Explainability Report

SHAP Attribution & Transparent Decisioning · Team Envirex

1. Design Principle: No Black-Box Scoring

Every risk score produced by PRITHVEX is required to come with an evidence trail. The platform's explicit design goal — stated end to end in the uncertainty report workflow — is explainable calibrated intelligence, with no black-box scoring at any stage of the pipeline. This document describes the SHAP-based attribution layer that delivers that guarantee.

2. SHAP Attribution Layer

PRITHVEX uses SHAP (SHapley Additive exPlanations) to decompose every risk score into signed, additive contributions from individual input features. SHAP was deliberately chosen because it is an industry-standard framework already accepted by regulators in financial and healthcare AI applications, which makes it a defensible choice for supply chain risk decisions that may face similar scrutiny.

3. Representative Attribution Output

The table below shows a representative SHAP attribution snapshot generated during a live risk-scoring run, ranking the drivers behind an elevated risk score.

Driver	Contribution	Risk Category
Heavy Rainfall Index	+0.42	Environmental Risk
Typhoon Haikui Probability	+0.34	Climate Risk
Vietnam Port Congestion	+0.28	Logistics Risk
Supplier Shutdown Probability	+0.21	Supply Risk
Lead-Time Variance	+0.12	Operational Risk
Transport Delay	+0.11	Logistics Risk

In this scenario, a simulated Typhoon Haikui shock causes 12 SKUs to flip to high-risk status within three seconds of injection. The SHAP attribution layer immediately identifies Vietnam port congestion as the top driver of the elevated scores, which lets the recommendation engine target the right intervention — in this case an emergency stock build for the four most exposed SKUs — rather than a generic, undifferentiated alert.

4. From Attribution to Recommendation

The explainability layer feeds directly into the action playbook. Each AI-ranked mitigation strategy is generated and justified using the same SHAP evidence that produced the underlying risk score, so a procurement manager reviewing a recommendation can trace it back to the specific environmental, logistics, supply, or operational driver that triggered it.

- High impact — increase safety stock for the affected supplier or route.
- High impact — shift to an alternate qualified supplier.
- Medium impact — expedite shipment via an alternate routing.
- Low impact — delay low-priority orders where risk is already contained.

5. Reliability Diagrams

Alongside per-decision SHAP trails, PRITHVEX renders a reliability diagram comparing predicted confidence against actual outcomes (see the companion calibration report for the Expected Calibration Error methodology behind this diagram). Together, the SHAP trail and the reliability diagram give a reviewer two independent, complementary views into why the model said what it said: which features drove this specific score, and how trustworthy the model's confidence has historically been.

6. Network-Level Explainability

Explainability is not limited to single-node scores. The Tier-1/2/3 network intelligence layer exposes which upstream node in a Monte Carlo shock-propagation run contributed most to a downstream risk increase, so a risk team can see not just that a score changed but which part of the supplier network caused it to change.

7. Governance Benefits

- Auditability — every alert and recommendation can be traced to a specific, ranked set of contributing factors.
- Regulatory readiness — SHAP-based attribution aligns with explainability expectations already established in financial and healthcare AI.
- Trust — users are shown evidence rather than asked to accept a bare numeric score.
- Faster triage — risk teams can act on the top driver immediately rather than investigating from scratch.